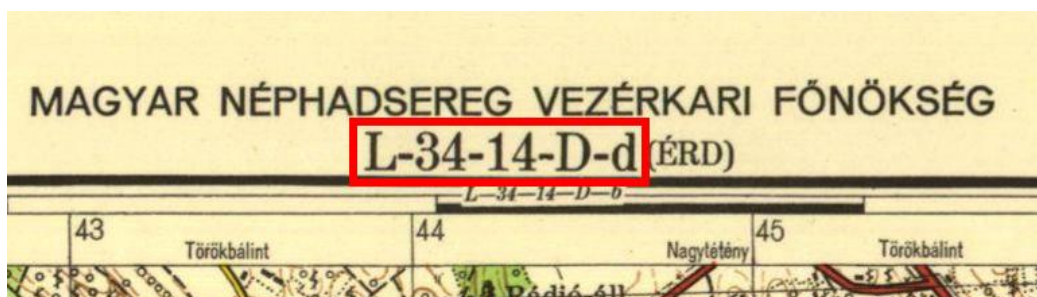




and to read the sheet identifier:



The sheet identifier consists of letters and digits separated by hyphens, following this scheme:
 {A-Z}-{1-60}-{1-144}-{A-D}-{a-d}

You can find randomly chosen 100 map sheets in test data, as well as a CSV file containing the corner coordinates of 10 of them.

Your program should:

- ask the user to enter the path where the scanned files are (such as "d:/data/maps/").
- iterate over the files at the given path. Find the corner coordinates and read the sheet identifier.
- create a text file in which each map sheet is represented by a line containing its filename, the x,y coordinates of the corners in (upper left, upper right, lower right, lower left) order, and the recognised sheet ID. Example:

```
L-33-009-D-d.jpg,308,279,2681,293,2670,2633,289,2620,L-33-9-D-d
L-33-010-C-b.jpg,303,295,2661,303,2647,2613,281,2630,L-33-10-C-b
L-33-010-C-c.jpg,367,288,2736,294,2727,2635,350,2629,L-33-10-C-c
```

Although the filenames of the test images contain the sheet IDs, it is only to help evaluating your solution. The final dataset won't be using this naming convention.

The submitted programs will be tested on the full map set, consisting of more than 1000 maps. Your final grade depends on:

- the accuracy of corner detection,
- the accuracy of sheet ID detection,
- the speed of the program.

The minimum requirements are:

- not more than 15 pixel error in corner detection for 85% of the sheets,
- at least 55% of the sheet identifiers is read without error,
- average sheet processing time is not more than 20 seconds.

Planned grading:

Up to 10 points for corner detection accuracy, up to another 10 points for sheet ID reading accuracy:

Points	Corner detection error
2	<15px for 85%
4	<15px for 90%
6	<15px for 92%
8	<10px for 92%
10	<10px for 95%

Points	Sheet ID reading accuracy
2	>50%
4	>60%
6	>65%
8	>70%
10	>75%

Structure of the sheet frames:

[sheet_frame_dimensions.jpg \(https://canvas.elte.hu/courses/61560/files/4204979/download?wrap=1\)](https://canvas.elte.hu/courses/61560/files/4204979/download?wrap=1)